

# Rishav Gupta

Ph.D. Student, Department of Computer Science and Electrical Engineering  
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## RESEARCH INTERESTS

Mobile health sensing and mobile systems, with an emphasis on acoustic and ultrasonic sensing, digital signal processing, and machine learning for cuff-free cardiovascular monitoring on commodity smartphones.

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## EDUCATION

- University of Maryland, Baltimore County** Jan. 2025 – Present  
*Ph.D. in Computer Science*  
• Advisor: [Dr. Dong Li](#), [Future Sensing and Interaction \(FSI\) Lab](#).  
**Baltimore, MD**
- University of Maryland, Baltimore County** Jan. 2025 – Dec. 2025  
*M.S. in Computer Science* (earned en route to the Ph.D.)  
• GPA: 4.00/4.00. Coursework: Design and Analysis of Algorithms, Machine Learning, Neural Networks.  
**Baltimore, MD**
- Manipal Institute of Technology** Jun. 2015 – May 2019  
*B.Tech. in Information Technology*  
• Ranked 8th out of 132 students in the graduating class.  
**Manipal, India**

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## PUBLICATIONS

Author names in **bold** indicate the CV holder.

- [1] **Rishav Gupta**, Alyssa N. Maguina, Sithumina Weeraratna, Janelle P. Clark, and Dong Li. “WiP: Smartphone-based Acoustic Tissue Characterization for Personalized Body-contact Systems.” In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Work-in-Progress Track. ACM, 2026. To appear.
- [2] **Rishav Gupta**, Shirui Cao, Jie Xiong, Sunghoon Ivan Lee, Jeremy Gummesson, and Dong Li. “Demo: Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones.” In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Demo Track. ACM, 2026. To appear.
- [3] Renzhi Hao, Shirui Cao, Yizhu Wen, **Rishav Gupta**, Dharani Nadendla, Mehran Kafai, Hanqing Guo, and Dong Li. “Ultrasound Watermark: Real-time Acoustic Watermarking against Voice Scams on Smartphones.” In *Proceedings of the 9th ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP '26)*, co-located with ACM/IEEE SEC. IEEE, 2026. To appear.
- [4] Shirui Cao, Jie Xiong, **Rishav Gupta**, Sunghoon Ivan Lee, Jeremy Gummesson, and Dong Li. “Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones.” In *Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services (MobiSys '26)*, Emerging Ideas Track. ACM, 2026. [\[Paper\]](#) [\[Code\]](#)
- [5] Renzhi Hao, Shirui Cao, Yizhu Wen, **Rishav Gupta**, Dharani Nadendla, Mehran Kafai, Hanqing Guo, and Dong Li. “Ultrasound Watermark: Real-time Acoustic Watermarking against Voice Scams on Smartphones.” *Under review*, ACM MobiCom 2027. Extended version of [3].

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## HONORS AND AWARDS

- Computing for Social Good Special Award**, COEIT Research Day, UMBC May 2026  
• One of only two awards presented across 114 posters; conferred by Dean Jeanne M. van Briesen.
- First Place**, Maryland NEW VENTURE™ Program Pitch Night, bwtech@UMBC May 2026  
• Awarded for *PocketCardio*, a smartphone-based cardiovascular screening venture.
- Best Poster Award**, CSEE Research Day, UMBC Apr. 2026
- Best Poster Award**, COEIT Research Day, UMBC May 2025

## PROFESSIONAL EXPERIENCE

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### University of Maryland, Baltimore County

Jan. 2025 – Present

*Graduate Research Assistant, Future Sensing and Interaction (FSI) Lab*

*Baltimore, MD*

### Gojek (GoTo Group)

Feb. 2022 – Sep. 2024

*Senior iOS Engineer*

*Bengaluru, India*

- Reduced annual platform cost by \$3M by replacing a third-party analytics SDK (CleverTap) with an in-house, open-source tool at full feature parity.
- Cut development time by 20% across the engineering team by automating the bi-weekly CI/CD release pipeline in Python and Shell on GitLab.
- Doubled QA throughput with a real-time Event Visualizer for verifying analytics events in situ, and cut roughly 3 MB from the app binary through SwiftSyntax AST-based dead-code detection.

### Nuclei

Jul. 2019 – Feb. 2022

*Senior iOS Developer and Team Lead*

*Bengaluru, India*

- Founding iOS developer of FedMobile (Federal Bank); implemented 60% of shipped features, including Payments, Investments, Loans, Onboarding, and Card Management.
- Promoted to Team Lead; led four iOS developers and closed a four-month delivery gap between the iOS and Android platforms.

### Hera doo

Jun. – Jul. 2018

*Android Developer Intern*

*Mostar, Bosnia*

- Built an end-to-end Android sales-order application from scratch, increasing field sales by 25%.

## SERVICE AND LEADERSHIP

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**Chair**, Graduate Assistant Advisory Committee (GAAC), UMBC Graduate Student Association

2026 – 2027

**HR Head**, ISTE Students' Chapter, Manipal Institute of Technology

2017 – 2018

**Co-Event Head** (Acumen), TechTatva '17, Manipal Institute of Technology

Oct. 2017

**Management Committee Member**, ISTE Students' Chapter, Manipal Institute of Technology

2016 – 2017

## TECHNICAL SKILLS

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**Languages** Swift, Python, Java, Shell, SQL, HTML/CSS

**Methods** Acoustic and ultrasonic signal processing, physiological sensing, tissue biomechanics, embedded audio, machine learning

**Tools** Xcode, SwiftUI, Android Studio, Git, GitLab CI/CD, L<sup>A</sup>T<sub>E</sub>X

## REFERENCES

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### Dr. Dong Li

Assistant Professor, Department of Computer Science and Electrical Engineering  
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